AMIRREZA KHOSHNIAT

Bojnord, Iran | +98 936 229 1523 | amirreza.kh76@gmail.com | LinkedIn: amirreza-khoshniat

RESEARCH PROFILE

MSc graduate in Medical Virology with hands-on experience in H9N2 avian influenza research. Skilled in virological assays, molecular methods, and sequence analysis. Master's research focused on the PB1-F2 and PA-X accessory proteins. Seeking doctoral training in influenza evolution, viral pathogenesis, and virus-host interactions.

EDUCATION & ACADEMIC DISTINCTIONS

MSc, Medical Virology Sep 2022 – Feb 2026

Razi Vaccine and Serum Research Institute, Karaj, Iran | GPA: 19.00/20 (Iranian scale)
National rank 16, Iran's 2022 Master's entrance examination; tuition-free admission.

BSc, Medical Laboratory Sciences Jan 2015 – Feb 2019

North Khorasan University of Medical Sciences, Bojnord, Iran
GPA: 16.00/20 (Iranian scale); tuition-free admission.

RESEARCH EXPERIENCE

Research Project Lead – Hepatitis B Education & Prevention Oct 2026

Islamic Azad University, Bojnourd Branch, Iran
Organized a prevention and occupational-exposure workshop for medical, nursing, and midwifery students; designed pre/post questionnaires. Questionnaire results and analysis pending.

Master's Thesis Researcher – Molecular Virology Feb 2024 – Feb 2026

Department of Avian Disease Research and Diagnostics, Razi Vaccine and Serum Research Institute, Karaj, Iran
Characterized PB1-F2 and PA-X in 23 Iranian H9N2 isolates (2017–2023), comparing PB1-F2 with 28 GenBank sequences. Identified predicted 57-amino-acid PB1-F2 truncation in 21/23 isolates (91.3%), with relative PA-X conservation.

Research Assistant – Experimental Pharmacology Feb 2017 – Feb 2018

North Khorasan University of Medical Sciences, Bojnord, Iran
Contributed to a rat-colitis study through tissue processing, histopathology, oxidative-stress assays, and statistical analysis.

Research training & scholarly contributions:

2026: Contributed chapters to *Virus-Associated Cancers: Molecular Pathogenesis and Genetic Susceptibility*; ISBN 978-622-367-988-9.

2025: PCR, Real-Time PCR & Data Analysis (Dec); Introduction to Bioinformatics (Jul), Iranian Union of Biologists and Iranian Academy of Bioinformatics.

TECHNICAL SKILLS

Virology: Embryonated-egg influenza propagation; HA/HI assays; viral titration; plaque assays; TCID50; BSL-2 practices.

Molecular biology: Viral RNA extraction; cDNA synthesis; PCR, RT-PCR and qPCR/RT-qPCR; agarose gel electrophoresis; primer design; Sanger chromatogram review and sequence editing.

Cell culture & immunology: Mammalian cell culture; ELISA.

Bioinformatics: Sequence alignment, phylogenetics and bootstrap analysis; Chromas, BioEdit, MAFFT, MEGA12, BLAST and GenBank; protein prediction/modeling with PSIPRED, SWISS-MODEL and PyMOL.

Experimental & statistical methods: Animal handling, tissue collection and histology; MDA, thiol, SOD/GPx assays; spectrophotometry; SPSS, GraphPad Prism, ANOVA and Tukey tests.

PUBLICATIONS & MANUSCRIPTS

Published journal articles

Charoghdoozi, K., **Khoshniat, A.**, et al. (2025). Matrine and oxymatrine; natural products with antiviral potentials. *Journal of Functional Foods*, 133, 107019. [Review article] doi: 10.1016/j.jff.2025.107019

Keshavarzi, Z., Safari, F., Alipour, B., **Khoshniat, A.**, et al. (2022). Antioxidant effects of methanol extract of *Dracocephalum kotschyi* in acetic acid induced rat colitis model. *Journal of Advances in Medical and Biomedical Research*, 30(1). [Original research] doi: 10.30699/jambs.30.e56821

Unpublished work

Khoshniat, A., et al. Characterization of Different Isoforms of PB1-F2 and PA-X in Iranian H9N2 Influenza Viruses. *Original research; in preparation.*

Khoshniat, A., et al. Application of Influenza Accessory Proteins, PA-X and PB1-F2 in Host Cells. *Review; completed, not submitted.*

Planned manuscript: Impact of an Educational Intervention on Knowledge, Attitudes, and Preventive Practices Regarding Hepatitis B Among Medical Sciences Students at Islamic Azad University, Bojnourd, Iran: A Quasi-Experimental Study. *Questionnaire results and data analysis pending.*

Additional reviews in preparation: Nanotechnology-enhanced oncolytic virotherapy (with Z. Azarash); HPV nanovaccines and the tumor immune microenvironment (with F. Oroojalian).

CONFERENCE PRESENTATIONS

2025 | Khoshniat, A., et al. "PA-X Accessory Protein: An Additional Contributor to the Pathogenicity of Avian Influenza Viruses." Oral presentation, 26th Iran's International Congress of Microbiology.

2023 | Azarash, Z., Moattari, A., & **Khoshniat, A.** "Nanoparticle-Based Flu Vaccine: The Past and Future." Poster, 4th Nanomedicine and Nanosafety Conference.

2018 | Khoshniat, A. "Prevalence of Urinary Tract Infections and Its Causative Agents in Imam Reza Hospital, Bojnurd, in 1396." Poster, 19th International & Iranian Congress of Microbiology.

CLINICAL EXPERIENCE

Clinical Laboratory Scientist

Jul 2024 – present | Dr. Taleghani Pathology Laboratory, Bojnord
Jul 2020 – Jul 2024 | Poursina Hospital, Bojnord
Sep 2019 – Jun 2020 | Sadaf Pathology Laboratory and Imam Ali Hospital, Chabahar
Mar – Jun 2019 | Pasteur Pathology Laboratory, Bojnord

Diagnostic testing and operation of hematology, biochemistry and immunoassay analyzers, including Roche cobas e 411 and IMMULITE 2000.

LANGUAGES

Persian: native | English: professional working proficiency

ACADEMIC REFEREES

Razi Vaccine and Serum Research Institute:

Dr. Abdolhamid Shoushtari (supervisor) | hamid1342ir@yahoo.com
Dr. Mohsen Bashashati (co-supervisor) | m.bashashati@rvsri.ac.ir
Dr. Mohammad Abdoshah (advisor) | m.abdoshah@rvsri.ac.ir

Dr. Amir Azimian | North Khorasan University of Medical Sciences
amir_azimian2003@yahoo.com